

HAGEN HAEUSSLER

hagen.haeussler@tum.de | [LinkedIn](#) | hagenhaeussler.com

EDUCATION

University of California, Berkeley – Computer Science & SWE Specialization

August 2025 – May 2026

- Fully funded (tuition fees & living costs) by a merit-based scholarship from the German Academic Scholarship Foundation
- CS 169L - Software Engineering Team Project: Learned to ship maintainable team software through code review, automated testing, feature flags, documentation & legacy-code integration in a Rails projects called [BJC Teacher Tracker](#), a UC Berkeley curriculum management systems
- ENGIN 183E - Technology Entrepreneurship: developed a full-stack [software solution for independent online tutors](#) after conducting over 50 customer interviews, implemented entire frontend & backend including Google Auth & Calendar integration & OpenAI API integration to bring AI to the tutoring domain, currently working with first beta users to scale the product
- EECS 151 - Intro. to Digital Design & Application Specific Integrated Circuits Laboratory: system architecture & programming
- CS 188 - Intro. to Artificial Intelligence: trained reinforcement learning model & leverage particle filtering algorithm to solve the game of Pacman, coded neural networks & MLP in Python to perform image recognition fine-tuning hyperparameters

Technical University of Munich (TUM) – B.Sc. Engineering Science

September 2023 – Present

- Broad, rigorous interdisciplinary engineering program (210 ECTS selective admission), Grade: 2.2/5.0, top 20% of class
- Awarded with the scholarships of the German National Academic Foundation (merit-based, top 0.5% German students) & the Federal German Scholarship "Deutschlandstipendium" (merit-based, top 1.2% of students in Germany)
- Took classes on: Computer Architecture, Semiconductors, CAD, UML, CMOS Architecture, RISC-V Architecture, Combinational & Sequential Logic, FSMs, Data Structures, Databases, Search Algorithms, SQL, C, Java & Python

Parker School, Hawaii – High School Exchange Year

October 2020 – June 2021

- GPA 4.0 | Android App Development, Computer Science, 3D Design | Head's List for "Outstanding Academic Performance"

Georg-Cantor-Gymnasium – High School

September 2014 – June 2023

- Abitur 1.0 (best possible grade, 2nd in class) | Focus: Mathematics & Natural Sciences (selective admission based on math & cognitive test)
- Awards: Membership in Deutsche Physikalische Gesellschaft, multiple State Math Olympiad & State Physics Olympiad awards
- Built & programmed my own LEGO robots, developed product websites using HTML only, designed & programmed traffic light system using FSM-based logic in C#, started developing my first own Android applications

ENGINEERING & RESEARCH PROJECTS

ASIC RISC-V & Cache Optimization Challenge – UC Berkeley

January 2026 – May 2026

- Designing direct-mapped cache to meet design specifications & optimize for area, energy & performance
- Achieved a 500% figure of merit increase & won the challenge (30+ participating teams)

Embedded Cyber-Physical Systems Lab – UC Berkeley

August 2025 – December 2025

- Designing embedded systems using Lingua Franca with C for timing, concurrency, PID-controlling & state-machine modeling
- Implementing low-level software interfacing with sensors/actuators on Pololu 3pi+ 2040 robot & Raspberry Pi Pico

Sim-to-Real Validation Research – UC Berkeley

August 2025 – December 2025

- Co-authored [research paper](#) on closing the Sim-to-Real gap using falsification-driven testing on Pololu 3pi+ robot
- Developed methodology to identify where robot simulations fail in reality – critical for safe deployment of autonomous systems

WORK EXPERIENCE & INITIATIVES

Tube Mill Control GmbH – Working Student

Mai 2026 - present

- Inline seam monitoring for welding systems, solely designed & developed the company's [website](#) end-to-end

Hack-Nation Venture Track

April 2026 - present

- Accepted into incubation program as 1 of 30 teams out of 1400 participants based on hackathon performance at the 5th Hack-Nation AI hackathon, built an agentic research automation web application leveraging frontier LLMs to boost scientific progress

Science x Entrepreneurship

March 2026 - present

- Empowering German deep tech talents in academia to launch their own startup, conducting field research & outreach

Porsche (Leipzig) – Servicing & Sensor Calibration

July 2022

- Reconfigured arc-welding monitoring systems for HKS Prozesstechnik GmbH

SONOTEC GmbH – Internship & Shadowing Program

May 2019

- Built & programmed a non-contact ultrasonic sensor for industrial tank quality monitoring

KSB SE & Co. KGaA – Industrial Mechanics Internship

October – November 2018

- Manufactured high-precision machine components using industrial lathes & milling machines

PERSONAL & SKILLS

- Judo Black Belt (1. Dan):** Central German Judo Championships U21 - 1st place
- Leadership:** 8x class president, 2x school president, founded school YouTube channel
- Munich Music Association (AGVM):** manage social media reaching 3000+ followers (Insta: @agvmuenchen, Facebook: Socialmedia Agv)